1 Solution — GIAM 3.2 Problem 5

1.1 Problem

Show that if x is a positive real number, then $x + \dfrac{1}{x} \geq 2$.

1.2 Finding the Proof — Work Backwards

The hypothesis "$x > 0$" is not much to push forwards with, so start at the conclusion and simplify until something obviously true appears. Crucially, **because $x > 0$ every step is reversible**:

$$\begin{aligned} x + \tfrac{1}{x} \geq 2 &\iff x^2 + 1 \geq 2x && \text{multiply by } x > 0 \\ &\iff x^2 - 2x + 1 \geq 0 \\ &\iff (x-1)^2 \geq 0 && \text{always true} \end{aligned}$$

1.3 The Proof

> ***Proof.*** *Let x be a positive real number. Since the square of a real number is non-negative,*
>
> $$(x-1)^2 \geq 0 .$$
>
> *Expanding, $x^2 - 2x + 1 \geq 0$, so $x^2 + 1 \geq 2x$. Dividing both sides by x, which is positive and therefore preserves the inequality,*
>
> $$x + \frac{1}{x} \geq 2 . \blacksquare$$

Verified on $20{,}000$ random positive reals.

1.4 The Picture

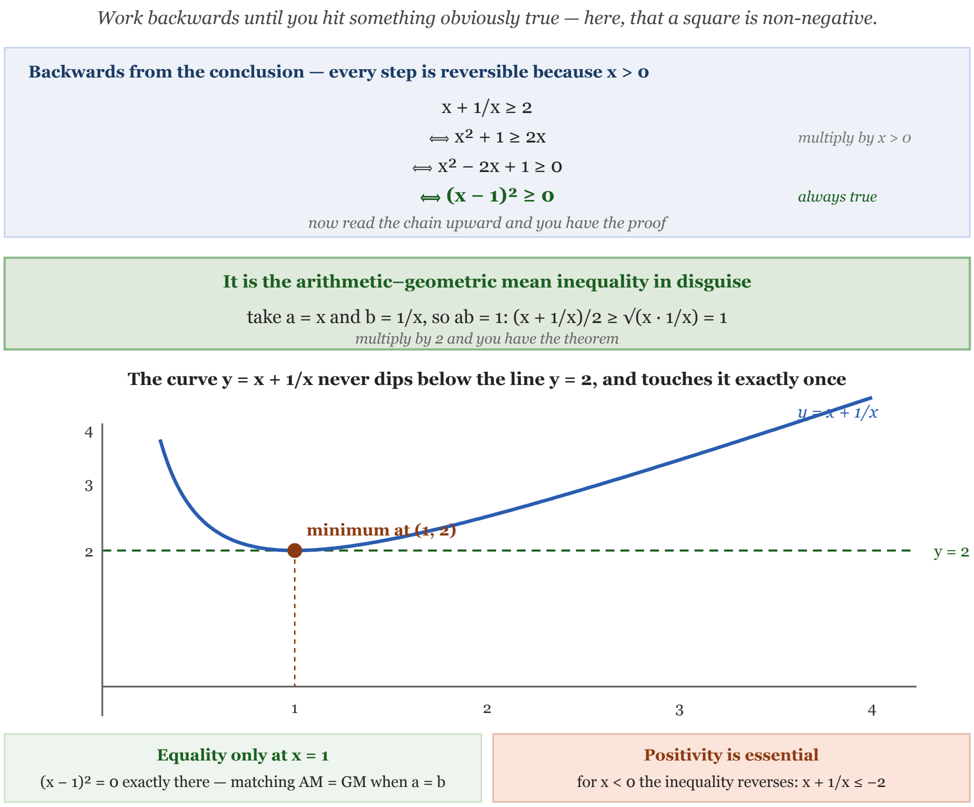

Figure 1.1: Top panel, backwards from the conclusion, every step reversible because x is positive: x plus one over x at least 2 is equivalent to x squared plus 1 at least 2x, multiplying by x positive; equivalent to x squared minus 2x plus 1 at least zero; equivalent to (x minus 1) squared at least zero, which is always true. Reading the chain upward gives the proof. A green panel notes it is the arithmetic-geometric mean inequality in disguise: taking a equals x and b equals one over x, so their product is 1, AM-GM gives (x plus one over x) over 2 at least the square root of 1, which is 1; multiplying by 2 gives the theorem. Below, a graph of the curve y equals x plus one over x on the interval from about 0.2 to 4, plotted above a dashed horizontal line at y equals 2. The curve descends steeply from the left, touches the dashed line exactly once at the point (1, 2) marked with a dot, then rises gently. Two closing panels: equality holds only at x equals 1, where (x minus 1) squared is zero, matching the AM equals GM condition a equals b; and positivity is essential, since for x negative the inequality reverses to x plus one over x at most minus 2.

1.5 Remark — Writing the Chain Backwards Is the Whole Technique

This is the section's method in its purest form. Discovering the proof and *writing* it are opposite journeys: you find $(x-1)^2 \geq 0$ by descending from the conclusion, then present the argument by climbing back up from it.

The section warns that this is only legitimate if the steps are **biconditional**. Here they are, and it is worth checking which:

step	reversible?	why
multiply by x	yes	$x > 0$, so no sign flip and no information lost
move $2x$ across	yes	adding the same quantity to both sides
factor $x^2 - 2x + 1 = (x-1)^2$	yes	an algebraic identity

Table 1.1

Had any step been one-directional, the backwards chain would have been a *discovery* only, and the forwards write-up would need separate justification.

1.6 Remark — This Is AM–GM With $a = x$, $b = 1/x$

The section's headline theorem is $\frac{a+b}{2} \geq \sqrt{ab}$ for $a, b \geq 0$. Setting $a = x$ and $b = 1/x$ gives $ab = 1$, so

$$\frac{x + 1/x}{2} \geq \sqrt{1} = 1 \qquad \Longrightarrow \qquad x + \frac{1}{x} \geq 2.$$

That is the entire problem, in one substitution. It is worth seeing both routes: the direct $(x-1)^2 \geq 0$ argument is self-contained, while the AM–GM route explains *why* the answer is 2 — because a number and its reciprocal always have geometric mean exactly 1, so their arithmetic mean cannot be less.

Notice too that the two proofs are the same proof. AM–GM itself is proved (in this section) by reducing to $(\sqrt{a} - \sqrt{b})^2 \geq 0$, and with $a = x$, $b = 1/x$ that is $(\sqrt{x} - 1/\sqrt{x})^2 \geq 0$ — which expands to the same statement.

1.7 Remark — Equality Holds at Exactly One Point

The proof gives more than the inequality: since $(x-1)^2 = 0$ only when $x = 1$, the inequality is **strict** for every positive $x \neq 1$, with equality at $x = 1$ alone.

That matches AM–GM's equality condition $a = b$, which here reads $x = 1/x$ — and for positive x that forces $x = 1$. Geometrically, the curve $y = x + 1/x$ has its minimum value 2 at $x = 1$ and rises on both sides, touching the line $y = 2$ exactly once. The chain of biconditionals is what lets us read off the equality case for free: the conclusion is tight precisely when the bottom line is.

1.8 Remark — Positivity Is Essential, Not Cosmetic

The hypothesis is used once, to divide by x without flipping the inequality. And it is genuinely needed — the statement is **false** for negative x:

$$x = -1: \quad -1 + \frac{1}{-1} = -2 \ngeq 2.$$

In fact for every $x < 0$ the inequality reverses completely: $x + 1/x \leq -2$, verified on 5000 random negatives. The reason is visible in the algebra — for $x < 0$, dividing $x^2 + 1 \geq 2x$ by x flips the relation, giving $x + 1/x \leq 2$; combined with the mirror argument via $(x+1)^2 \geq 0$ one gets the sharp bound -2.

So the full picture is $|x + 1/x| \geq 2$ for all $x \neq 0$, with the sign of x deciding which side.

1.9 Remark — Why $(x-1)^2$ and Not Something Else

The step that looks like a rabbit from a hat is recognising $x^2 - 2x + 1$ as a perfect square. But nothing was guessed: **completing the square** is the systematic way to decide whether a quadratic is always non-negative.

$$x^2 - 2x + 1 = (x - 1)^2 + 0,$$

and a quadratic $x^2 + bx + c$ is non-negative everywhere exactly when its discriminant $b^2 - 4c \leq 0$ — here $4 - 4 = 0$, the borderline case, which is precisely why equality is achieved rather than the inequality being strict everywhere. This connects back to Problem 2: the discriminant governs whether a parabola touches, crosses, or misses the axis, and a *zero* discriminant is the tangency case that produces exactly one equality point.